

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks:50

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

*I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

TECHNICAL PRESENTATION

1. From Caesar to Vigenère: Evolution of Substitution Ciphers”
2. “Substitution vs Transposition: Which Provides Better Security?”
3. “Security Attacks, Threats, Risks and Their Impact on Security Goals”
4. “Steganography vs Cryptography: Hidden Security in the Digital World”
5. “Role of Number Theory in Cryptography: From Euclid to Euler”

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



SIMATSENGINEERING

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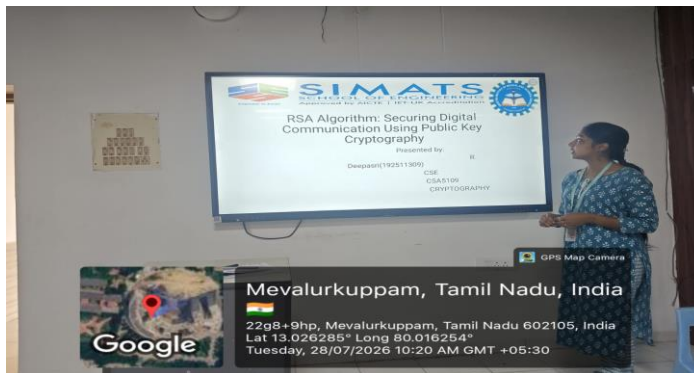
Course Code: CSA5109

Slot:D

Course Name: Cryptography

Course Faculty: Dr. Supriya

Project title: RSA algorithm: Securing Digital Communication using Public key cryptography



Project Description:

- The "Securing Digital Communication Using Public Key Cryptography" project is designed to ensure the safe exchange of information over digital networks. With the rapid growth of the internet and online services, protecting sensitive data such as passwords, financial information, personal details, and confidential messages has become essential. This project uses Public Key Cryptography, a secure encryption technique that employs two different keys: a public key for encrypting data and a private key for decrypting it. Since the private key is known only to the intended recipient, unauthorized users cannot access the original information even if they intercept the encrypted message.
- The project mainly demonstrates the working of the RSA (Rivest–Shamir–Adleman) algorithm, one of the most widely used public key cryptographic algorithms. RSA is used to encrypt messages, securely exchange keys, and create digital signatures for authentication. The system also ensures confidentiality, authentication, data integrity, and non-repudiation, which are the key principles of secure communication. Confidentiality ensures that only authorized users can read the message, authentication verifies the identity of the sender, data integrity confirms that the message has not been modified during transmission, and non-repudiation prevents the sender from denying that they sent the message.

Student Signature

Guide Signature